

CSE 3105 / CSE 3137

OBJECT-ORIENTED ANALYSIS AND DESIGN

FALL 2025

COURSE PROJECT: *<SMART CITY CLEAN>*

Requirements Analysis Document

Group 5

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6 November 2025

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1 Introduction

The main purpose of this document is to present the Requirements Analysis Document (RAD) for the "Smart City Clean - Smart Waste and Cleaning Management System" project we have developed. The primary function of the system is to be an intelligent management application designed to monitor and manage waste collection and urban cleanliness. The motivation behind the project's development stems from eliminating the unpleasant odors, visual pollution, and health risks caused by frequently overflowing waste bins in the current (Current) system, which are due to population density and inefficient fixed schedules. In this context, the fundamental objective of the project is to optimize waste collection schedules using IoT sensor data and citizen feedback, thereby reducing environmental pollution and improving city hygiene. The scope of the project is limited to the Requirements Analysis phase of this document, and the ultimate success criterion is the creation of a detailed software design, including UML diagrams (use case, class, sequence, and activity) and user interface mock-ups, that enhances cleanliness efficiency and sustainability.

2 Current System

1. Traditional Collection Systems:

• Fixed Schedule Route Application: Most municipalities still use traditional, static route planning methods that determine routes according to fixed days and times, regardless of the container fill level. This leads to unnecessary trips and resource waste.

• Dual Collection System and Waste Collection Centers: Within the scope of the Ministry of Environment, Urbanization and Climate Change's "Zero Waste Project," dual container systems (blue for recyclables, grey/dark grey for other waste) have been established in residences and public areas. Furthermore, citizens are encouraged to sort and bring their waste to glass bottle banks and Waste Collection Centers (AGM). However, this is not a system directly focused on reducing the cleaning staff’s workload or street pollution.

2. Smart Waste Management Pilot Projects:

• IoT Sensor Applications: In projects like the "Smart Container System" implemented by Pendik Municipality, ultrasonic sensors placed in containers measure fill levels and send alerts to the center when full. Some of these smart containers operate on solar energy and compress waste to increase capacity.

• GPS and GIS-Based Management: Some local governments, such as Çorum Municipality, use GPS data from vehicles and containers, integrated with a Geographical Information System (GIS) base, to ensure the efficient use, analysis, and planning of collection vehicles and containers. These systems provide route optimization and easy control at critical points.

• Citizen Interactive Incentive Systems: Projects named Environmentalist Neighbor Card, Recycling Card, or Waste Cashare found in major cities like Ankara, Istanbul, and Antalya. These systems offer incentives like points, cash, or cleaning products to citizens who bring their sorted waste.

3 Proposed System

The proposed system (Smart City Clean – Smart Waste and Cleaning Management System) is an integrated, data-driven platform designed to solve the existing problems of waste management and city cleanliness, increase operational efficiency, and minimize resource waste. The system will combine Internet of Things (IoT) sensors with citizen feedback to provide dynamic route optimization and real-time management.

3.1 Functional Requirements

ID	Requirement Description	Actor	Related Scenario
FR1.0	The System shall be able to monitor the fill level of each smart bin compartment (Metal, Paper, Plastic, Glass, Household Waste) in real-time using IoT sensors.	Smart Bin, System	Scenario 3
FR2.0	The System shall automatically send an alert (notification) to the Cleaning Staff and Municipal Manager when a compartment fill level exceeds a predefined threshold (e.g., 80%).	System, Cleaning Staff, Municipal Manager	Scenario 3
FR3.0	The System shall contain a route planning algorithm capable of optimizing collection routes based on collected fill level data (FR1.0) and citizen reports (FR6.0).	System	Scenario 3
FR4.0	The Cleaning Staff shall be able to view their assigned optimized route via a mobile application and confirm the completion of a bin emptying task to the System.	Cleaning Staff	Scenario 3
FR5.0	The Municipal Manager shall be able to monitor city cleanliness levels and all bin fill statuses via a real-time data visualization dashboard.	Municipal Manager	Scenario 3
FR6.0	The Citizen shall be able to report pollution or overflowing bins, including photo evidence, via a mobile application or web interface.	Citizen	Scenario 3 (Route input)
FR7.0	The Smart Bin shall be able to detect the type of waste being deposited (Metal, Paper, Plastic, etc.) and provide an audio/visual warning to prevent incorrect sorting.	Smart Bin	Scenario 2
FR8.0	The Smart Bin interface shall include a function to detect a dropped valuable item (e.g., wallet) and offer the Citizen an option to Alert Staff for item retrieval (Scenario 1).	Smart Bin, Citizen	Scenario 1
FR9.0	The System shall store timestamped records of all key interactions, including task completion, citizen reports, and incorrect waste attempts, for analysis and reporting purposes.	System	General Requirement

3.2 Nonfunctional Requirements

Performance

The route optimization algorithm must generate the most efficient route, considering all points to be collected, in less than 5 seconds.

Reliability

Automatic emergency alerts sent to the Cleaning Personnel when the fullness threshold (80%) is exceeded must be delivered successfully 99% of the time.

Usability

The task route interface in the Cleaning Personnel's mobile application must require a maximum of 2 clicks to update the status of a task to "Completed".

Operational

The sensors of the smart waste bins must sustain data transmission on battery power without external intervention for a minimum of 15 days.

Security

All sensor data, including waste bin fullness data and geographical location information, must be transmitted encrypted (e.g., via TLS/SSL) against unauthorized access.

Supportability

The mobile version of the application must support the last two major versions of the Android and iOS operating systems.

Interface

The Municipal Manager Dashboard must be displayed consistently and responsively across all modern web browsers (Chrome, Firefox, Edge).

3.3 System Models

3.3.1 Scenarios

Scenario 1

Scenario name: walletAccidentallyDroppedInBinAndRetrieved

Participating actor instances: citizen:Citizen, smartBin:SmartBin, system:System, staff:CLeaningStaff

Flow of events:

1. **citizen** attempts to throw paper waste into the "Paper" compartment of the **smartBin**.

2. As the waste is discarded, sensors (e.g., camera/volume) inside the **smartBin** detect that a wallet has also fallen in along with the waste.

3. **citizen** simultaneously notices that their wallet has fallen into the bin.

4. **smartBin** immediately displays an alert message on its screen: "You may have dropped a valuable item. Touch 'Alert Cleaning Staff' for retrieval."

5. **citizen** touches the "Alert Cleaning Staff" option on the screen and enters their contact information (name/phone).

6. **system** sends an SMS/Mobile App notification to the area's cleaning staff. This notification includes the **smartBin**'s location, the estimated type of the dropped item, and the **citizen**'s contact information.

7. **staff** receives the notification from the mobile app and immediately heads to the location.

8. When **staff** arrives at the location, they verify the citizen's identity.

9. After verification, **staff** unlocks the "Paper" compartment using special authorization via the mobile app (or the bin's interface).

10. **citizen** safely retrieves their wallet.

11. **staff** updates the system that the incident is resolved, and the system closes the case.

Scenario 2

Scenario name: incorrectWasteDisposalAndSystemAlert

Participating actor instances: citizen:Citizen, smartBin:SmartBin, system:System

Flow of events:

1. **citizen** approaches the "Plastic" compartment opening of the **smartBin**, intending to dispose of a glass bottle by mistake.

2. The **smartBin**'s waste classification system (camera/sensors) identifies the item's type as (Glass Bottle).

3. **smartBin** locks the "Plastic" compartment opening, preventing it from opening, and gives an audio/visual warning to the **citizen**: "This is glass waste. Please use the 'Glass' compartment."

4. Following the alert, **citizen** directs the glass bottle to the correct "Glass" compartment opening.

5. **smartBin** automatically opens the "Glass" compartment.

6. **citizen** disposes of the glass bottle into the "Glass" compartment.

7. **smartBin** updates the fill level of the (Glass) compartment.

8. **system** records the data related to this incorrect disposal attempt (Location, Waste Type, Error Type) to the data analysis platform for the Municipal Administrator.

Scenario 3

Scenario name: binFullnessAlertAndRouteOptimization

Participating actor instances: smartBin:SmartBin, system:System, admin:MunicipalAdministrator, staff:CleaningStaff

Flow of events:

1. **smartBin**'s sensors in the "Household Waste" compartment detect that the fill level has exceeded the %80 threshold.

2. **system** automatically sends an "urgent emptying alert" to the **admin** and the relevant **staff**.

3. **system** updates the status for this bin in the database to "Requires Emptying".

4. **admin** views the location and fill level of the bin on the rael-time dashboard.

5. **system** runs the route planning algorithm, taking into account the locations of all full bins (exceeding 80%) and other collection areas (e.g., citizen reports).

6. **system** generates the most efficient and optimized cleaning/emptying route.

7. **staff** receives the newly optimized task route (a sequenced list of stops and bins to be emptied) in their mobile application.

8. **staff**, following the route, empties the relevant bin and updates the task's status to "Completed" via the mobile application.

9. **smartBin**'s sensors register the emptying, and its fill level in the system is updated to 0%.

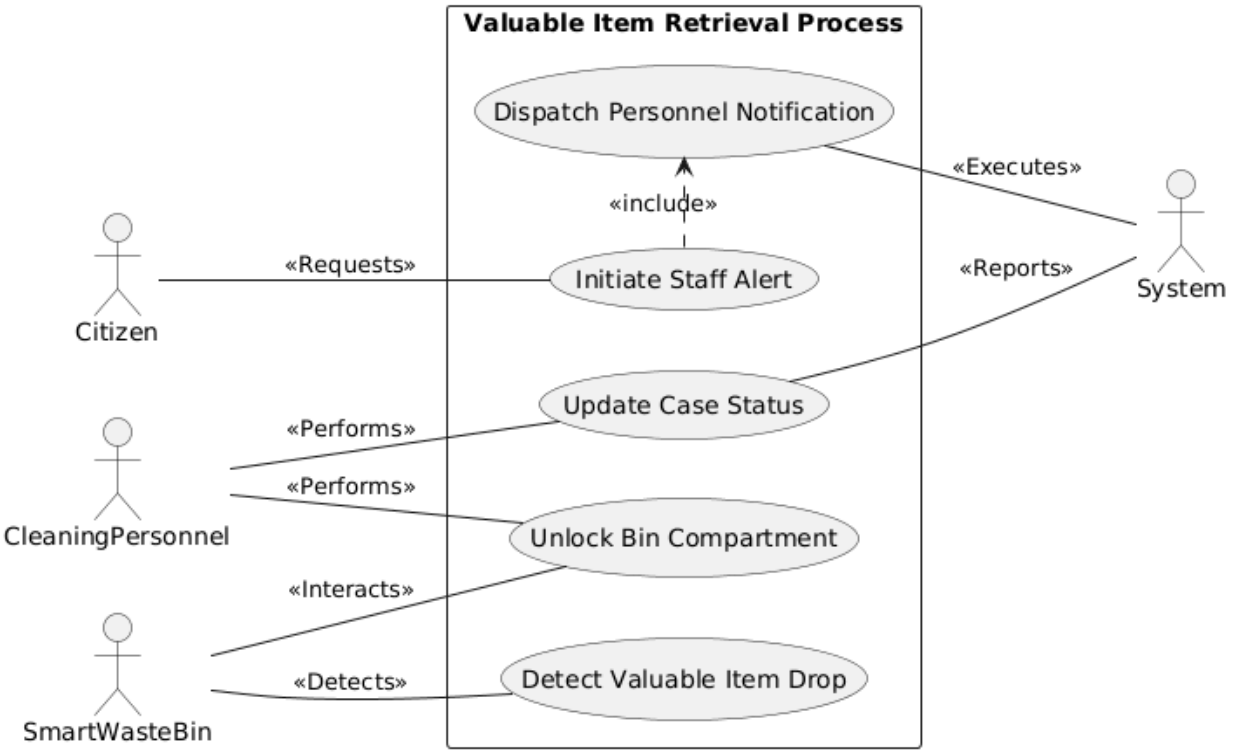
10. **system** updates the cleaning level indicators on the admin's dashboard.

3.3.2 Use Case Model

1. Scenario: Accidental Item Drop and Retrieval

Scenario Name: walletAccidentallyDroppedInBinAndRetrieved

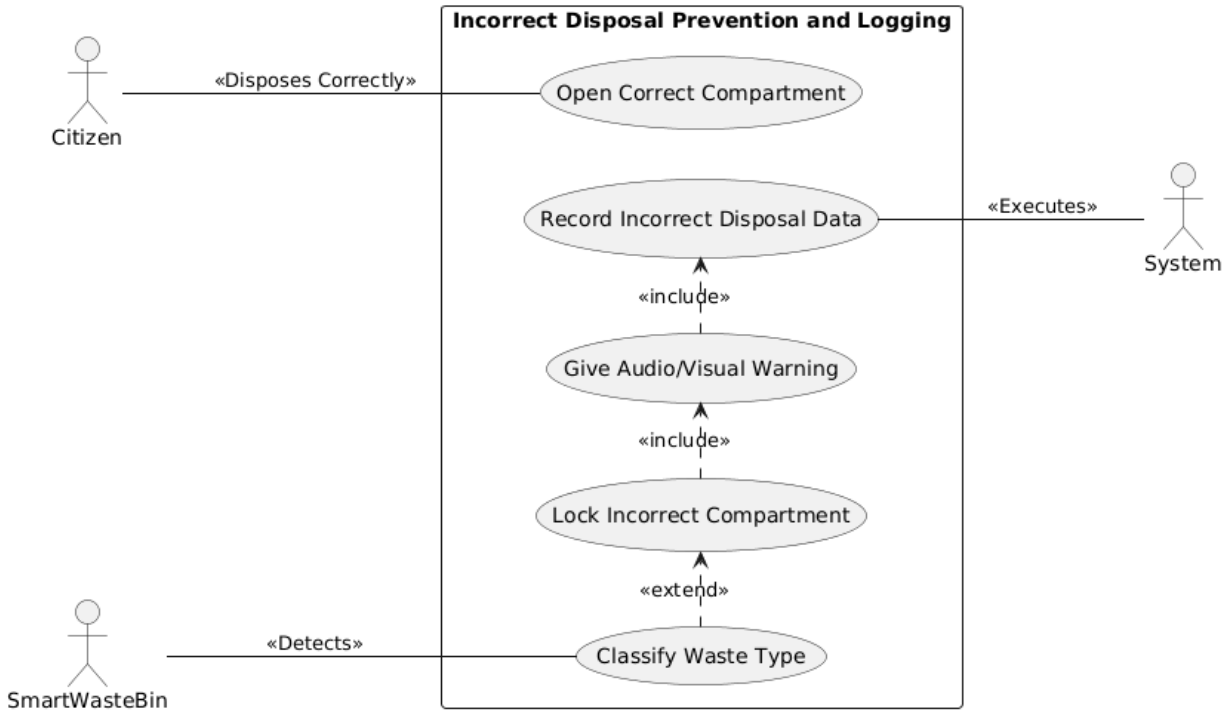
1.Name	Handle Accidental Item Drop and Staff Alert
2.Participating Actors	Primary: Smart Bin. Secondary: Citizen, System, Cleaning Personnel.
3.Entry Condition	1. The Citizen is disposing of waste. 2. The Smart Bin's internal sensors detect a sudden drop of a valuable item (e.g., wallet) in addition to the normal waste. 3. The related compartment is automatically put into a temporary locked state.
4.Exit Condition	1. The Citizen's contact information has been recorded (FR8.0). 2. The relevant Cleaning Personnel has been notified of the emergency location and details. 3. The item has been retrieved OR the incident has been formally logged as failed.
5.Flow of Events	5.1. Main Success Scenario (Item Retrieval): 1. The Smart Bin detects the dropped item and immediately displays an alert on its screen, prompting the Citizen to "Alert Staff" (FR8.0). 2. The Citizen selects the option and enters their contact information. 3. The Bin transmits the emergency alert and contact details to the System. 4. The System assigns the alert as an emergency task to the nearest Cleaning Personnel. 5. The Personnel arrives at the location, verifies the Citizen's identity, and uses their authorized code to open the compartment (via Open Bin Compartment action). 6. The Citizen retrieves the item, and the Personnel marks the task as "Completed." 5.2. Alternative Flow (AF): Delayed Personnel Response A.1. Personnel takes longer than 15 minutes to arrive. A.2. System: The System notifies the Municipal Manager of the delay for follow-up and resource management.
6.Special Requirements	1. The Smart Bin must include the Record Citizen Contact Info Use Case to ensure successful staff follow-up (FR8.0). 2. The process extends to the Open Bin Compartment action, which requires special Personnel authorization to execute.



2. Scenario: Incorrect Waste Disposal and System Alert

Scenario Name: incorrectWasteDisposalAndSystemAlert

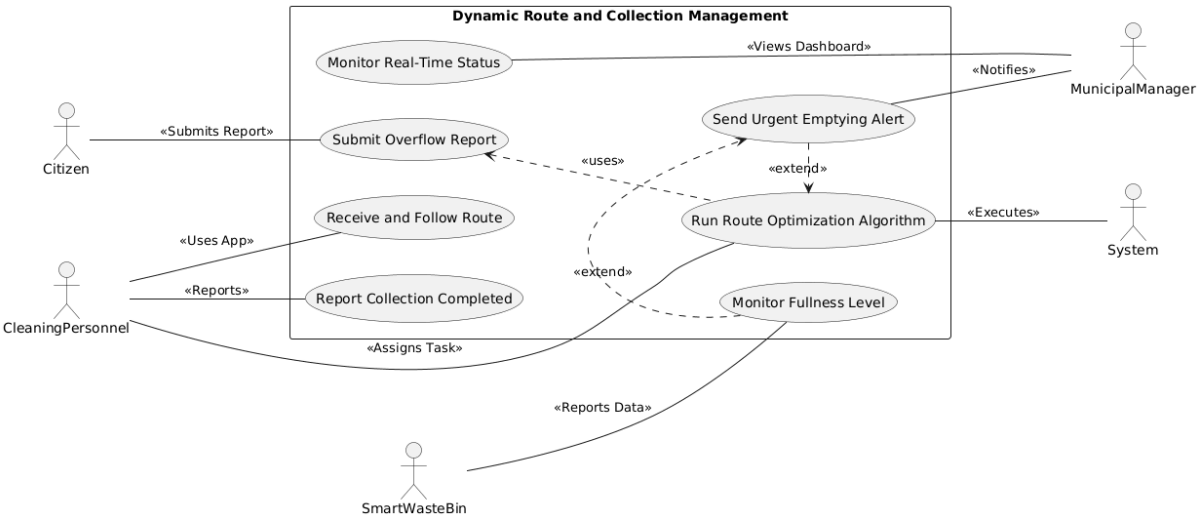
1.Name	Detect and Manage Incorrect Waste Disposal
2.Participating Actors	Primary: Smart Bin. Secondary: Citizen, System.
3.Entry Condition	1. A Citizen attempts to deposit waste into one of the bin's compartments. 2. The internal sensors/classification system are active and detecting the waste type.
4.Exit Condition	1. The correct compartment is opened, OR the incorrect compartment remains locked. 2. The Citizen has been informed/warned (audio/visual). 3. The incorrect waste attempt has been logged by the System (FR9.0).
5.Flow of Events	5.1. Main Success Scenario (Corrective Action): 1. The Citizen attempts to place a Glass Bottle into the Plastic compartment. 2. The Smart Bin classifies the item as "Glass" (FR7.0). 3. The Bin immediately locks the Plastic compartment door and issues a warning: "This is glass waste. Please use the 'Glass' compartment." (FR7.0). 4. The Citizen redirects the item to the correct "Glass" compartment. 5. The Bin unlocks and opens the Glass compartment for disposal. 6. The Bin logs the incorrect attempt and the successful final disposal with the System. 5.2. Alternative Flow (AF): Lock Compartment (<<extend>>) A.1. The Bin determines the incorrect disposal is severe (e.g., a toxic item). A.2. The Bin extends the main process by triggering the Lock Bin Compartment action, which permanently locks the compartment until maintenance is performed.
6.Special Requirements	1. The Smart Bin must include the Provide Audio/Visual Warning Use Case after detecting incorrect waste (FR7.0). 2. The waste classification system must operate with a minimum of 95% accuracy.



3. Scenario: Bin Fullness Alert and Route Optimization

Scenario Name: binFullnessAlertAndRouteOptimization

1.Name	Optimize Collection Route and Dispatch Task
2.Participating Actors	Primary: System (Central Management System). Secondary: Smart Bin, Cleaning Personnel, Municipal Manager, Citizen (indirectly via reports).
3.Entry Condition	1. A Smart Bin compartment has exceeded the fullnessThreshold (e.g., 80%). OR 2. A Citizen has submitted a pollution/overflow report. 3. The System is currently in Online status.
4.Exit Condition	1. The optimal collection route has been calculated and stored. 2. A new task containing the route is assigned to the relevant Cleaning Personnel. 3. The bin's fill level is updated (to 0% after completion) and operational data is logged.
5.Flow of Events	5.1. Main Success Scenario: 1. The Smart Bin sends a fullness alert to the System. 2. The System receives and processes the data, marking the bin location as required for collection. 3. The System runs the Route Optimization Algorithm (FR3.0) using all flagged locations as input. 4. System: The System dispatches the calculated route as a task to the designated Cleaning Personnel. 5. The Cleaning Personnel receives the task on their mobile app. 6. The Personnel empties the bin(s) and updates the task status to "Completed." 7. The System receives the "Completed" status, updates the bin's fill level to 0%, and logs the event (FR9.0). 5.2. Alternative Flow (AF): Optimization Failure (<<extend>>) A.1. The algorithm fails to find a route within the defined constraints (e.g., time, fuel). A.2. The System generates a partial route prioritizing the most critical bins (>95%). A.3. The System sends a high-priority intervention alert to the Municipal Manager. 5.3. Alternative Flow (AF): Triggered by Citizen Report B.1. A Citizen submits a pollution/overflow report (FR6.0). B.2. The System adds the reported location to the collection list and proceeds to Step 3 of the Main Success Scenario (Runs Route Optimization).
6.Special Requirements	1. The route optimization algorithm must generate the route in less than 5 seconds. 2. The Use Case must include the Monitor Fill Level Use Case to obtain the necessary input data. 3. The Use Case execution must include the Dispatch Collection Task action.



3.3.3 Object Model

<Dynamic models section documents the behavior of the object model in terms of state machine diagrams and sequence diagrams. Although this information is redundant with the use case model, dynamic models enable us to represent more precisely complex behaviors, including use cases involving many actors.>

Activity 5

3.3.4 Dynamic Models

<Dynamic models section documents the behavior of the object model in terms of state machine diagrams and sequence diagrams. Although this information is redundant with the use case model, dynamic models enable us to represent more precisely complex behaviors, including use cases involving many actors.>

Activity 5

3.3.5 User Interface Mock-ups

The visual mock-ups represent the integrated management interface designed for Municipal Managers and Cleaning Staff, focusing on real-time monitoring and operational efficiency.

Interface Components and Functionality:

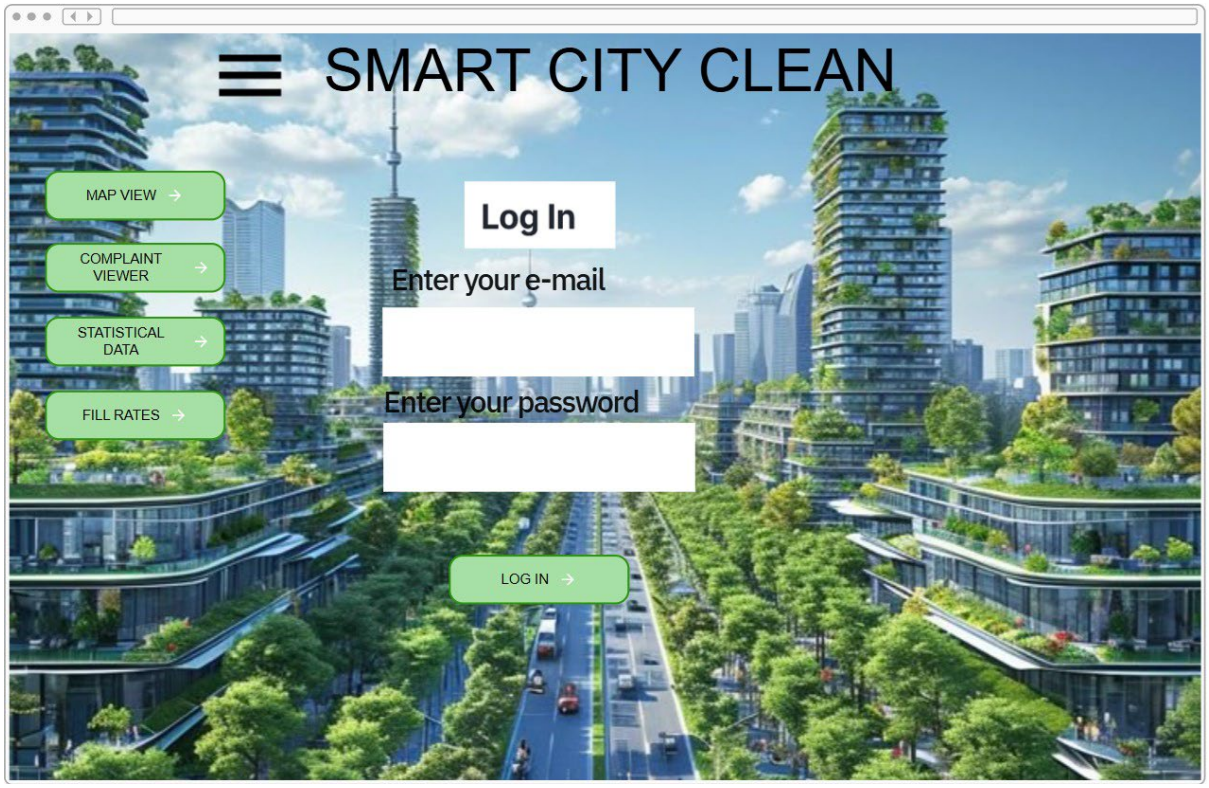
Navigation and Reporting: The primary navigation facilitates access to the Home Page, Reports, and Notifications/Alerts. The “Submit Report” button allows staff to manually report critical situations like overflow or environmental pollution encountered in the field. The “Notifications” list displays automatically generated system alerts, primarily triggered when fill levels exceed the 80% threshold (FR2.0). “Reports” offer detailed analytics on operational efficiency and past performance.

Real-Time Map and Optimization (UC-R01): The core of the application is the Map View, which displays the geographic location of all Smart Bins. Bins are color-coded based on their real-time fill level (e.g., Red for critical, >80%) (FR1.0). This visualization supports the Municipal Manager in monitoring the “City Cleanliness Status” (FR5.0). The map overlays the optimized route path determined by the Route Optimization Algorithm (FR3.0), replacing static collection schedules. This feature directly supports UC-R01 by enabling efficient task assignment (FR4.0).

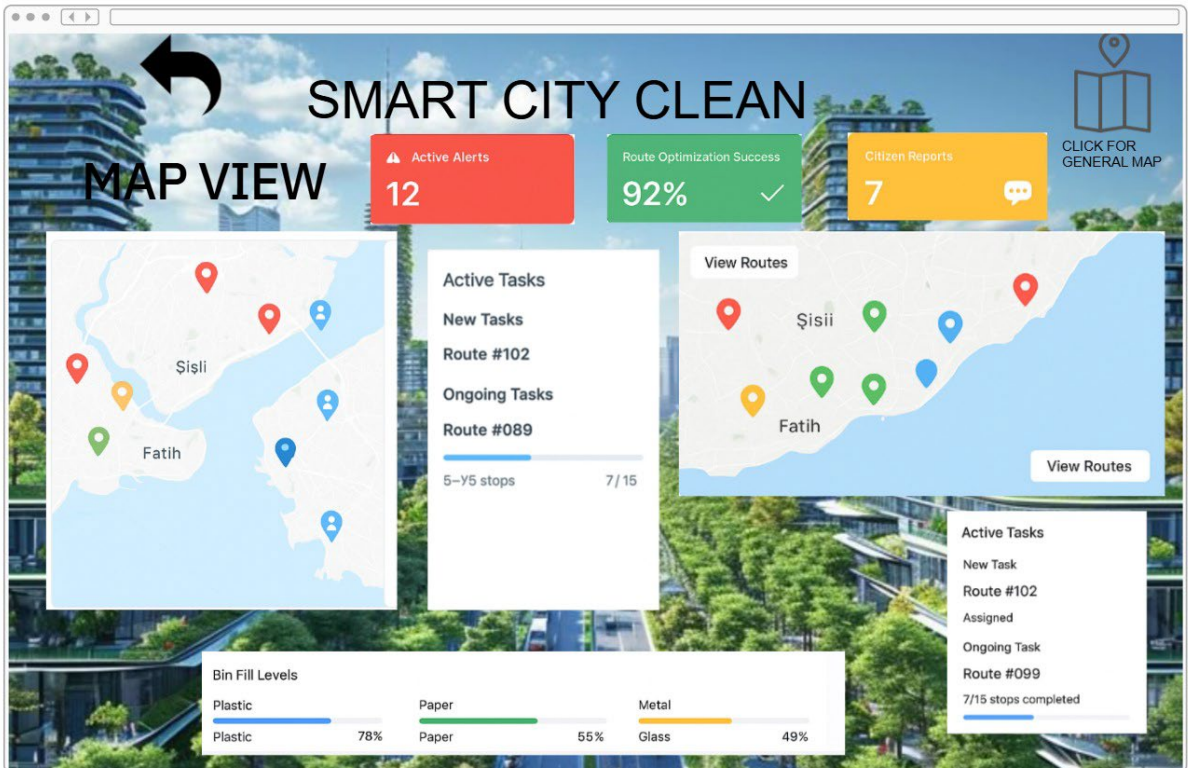
Bin Detail View (FR1.0): Tapping a bin icon reveals a detailed status screen, crucial for both actors. It shows the bin's unique identifier (binID) and location. Most importantly, it displays the specific fill percentages for each of the 5 compartments (Metal, Paper, Plastic, Glass, Household) (FR1.0). This ensures that staff knows exactly which compartment needs emptying.

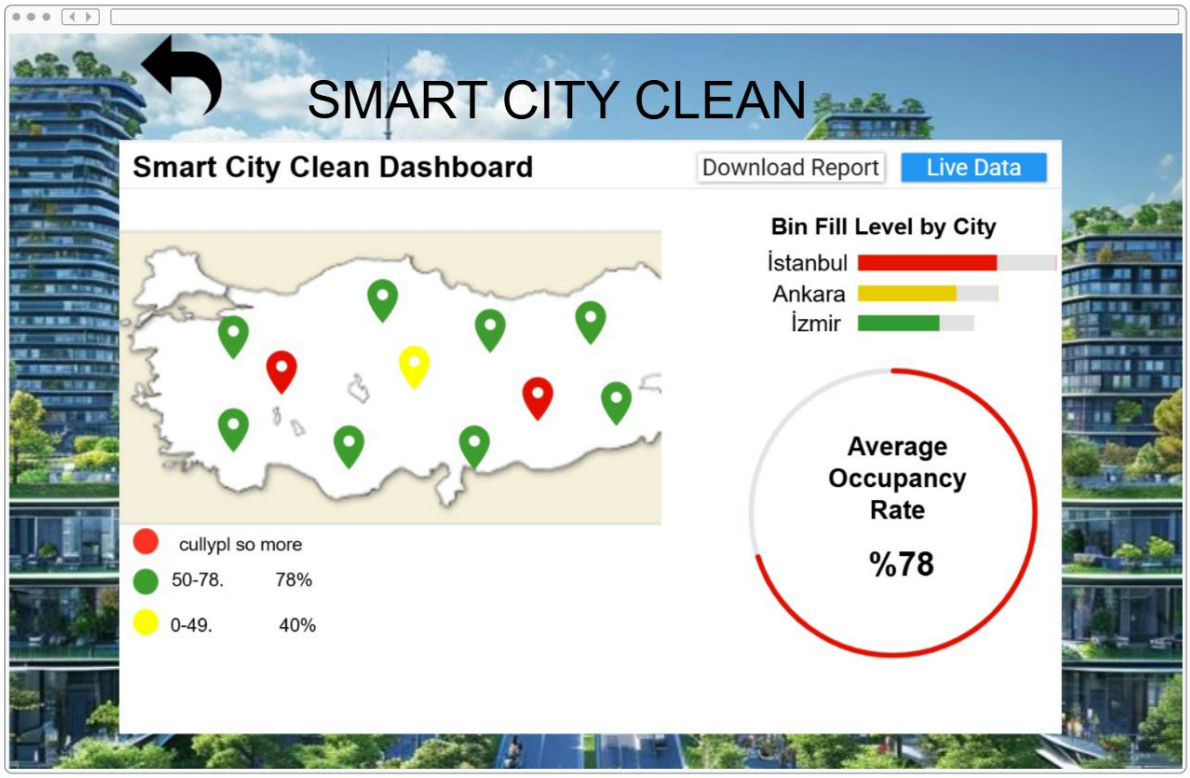
Operational and Analytical Tools: The interface includes a "Submit Report" function to record Citizen feedback on pollution. The overall system ensures that records of all key interactions, including task completion, are stored for analysis (FR9.0).

https://pr.to/DJZE2K/

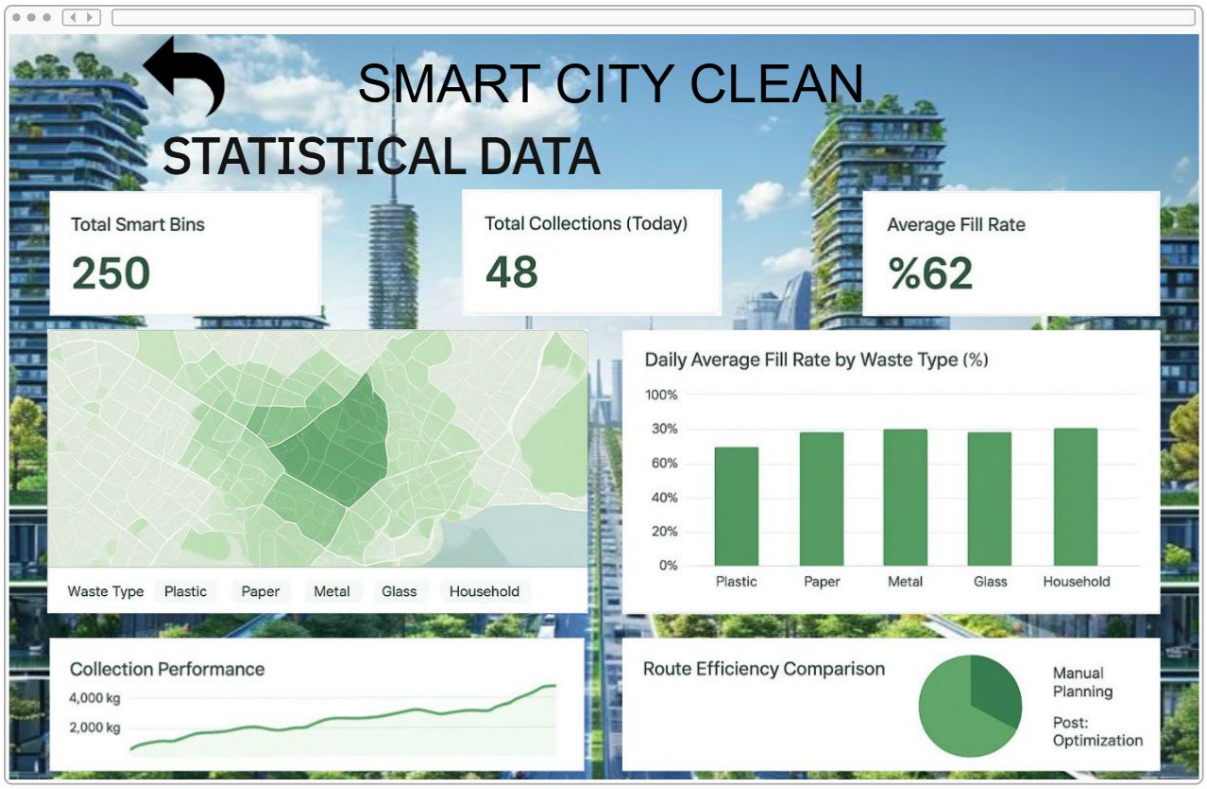





SMART CITY CLEAN
MAP VIEW
Active Alerts
12
Route Optimization Success
92%
Citizen Reports
7
CLICK FOR GENERAL MAP
Şişli
Fatih
Active Tasks
New Tasks
Route #102
Ongoing Tasks
Route #089
5–Y5 stops
7/15
View Routes
Şişii
Fatih
View Routes
Active Tasks
New Task
Route #102
Assigned
Ongoing Task
Route #099
7/15 stops completed
Bin Fill Levels
Plastic
Plastic 78%
Paper
Paper 55%
Metal
Glass 49%


SMART CITY CLEAN
Smart City Clean Dashboard
Download Report
Live Data
Bin Fill Level by City
İstanbul
Ankara
İzmir
cullypl so more
50-78. 78%
0-49. 40%
Average Occupancy Rate
%78


SMART CITY CLEAN
COMPLAINT VIEW
Smart Waste Management
Complaints
b0391b41-b8dd-41cf-b00f-b4b9daaac47c2
view report
The bin at the corner of Meclisi Street is overflowing.
Date: 6 April 2024
Reported by: Hasan Yilmaz
Location: Fatih, Istanbul


SMART CITY CLEAN
STATISTICAL DATA
Total Smart Bins
250
Total Collections (Today)
48
Average Fill Rate
%62
Waste Type
Plastic
Paper
Metal
Glass
Household
Daily Average Fill Rate by Waste Type (%)
100%
30%
60%
40%
20%
0%
Plastic
Paper
Metal
Glass
Household
Collection Performance
4,000 kg
2,000 kg
Route Efficiency Comparison
Manual Planning
Post: Optimization

The visual mock-ups represent the mobile application designed for Citizens, enabling active participation in waste management, reporting, and managing emergency scenarios. This application serves as the primary interface for Citizen actors (FR6.0, FR8.0).

Interface Components and Functionality:

Home/Map View: The main screen features a map displaying the Citizen’s current location and the location of Nearby Smart Bins. This allows the user to quickly find the nearest multi-compartment bin. The bins may be color-coded to provide a general indication of their fullness status, promoting user engagement with the system's core data (FR1.0).

Report an Issue (UC-G01 / FR6.0): This is a critical function for involving the public in cleanliness management. The dedicated report screen allows the Citizen to:

Submit Report: Initiate the process of logging a pollution or overflow incident.

Include Photo Evidence: Directly upload or take a photograph of the pollution or overflowing bin, which significantly enhances the report's credibility (FR6.0).

Specify Location: Confirm the exact location (auto-filled via GPS) or manually pinpoint the issue on the map.

This reporting mechanism directly feeds into the Route Optimization Algorithm (UC-R01), making Citizen input a key factor for task dispatch (FR3.0).

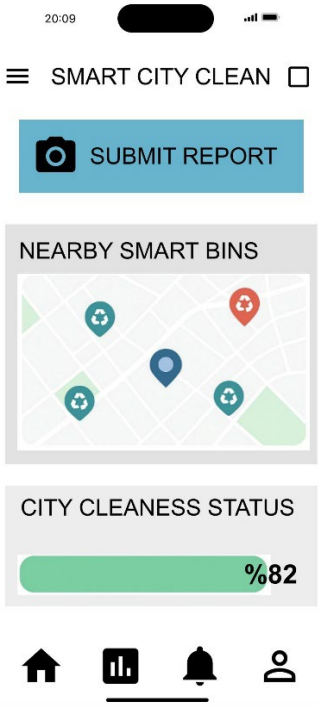
Notifications/Status Tracking: The interface includes sections for Notifications/Alerts. These alerts inform the user about the status of their submitted Reports (e.g., "Report received," "Personnel dispatched," "Issue resolved"). This closed-loop communication enhances user satisfaction.

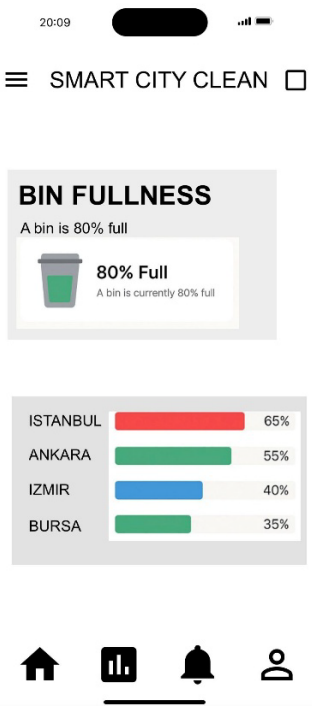
Emergency Feature (UC-E01 / FR8.0): Although not explicitly shown as a screen, the design implies an accessible method (either a dedicated app section or a prompt near the physical bin interface) for the Handle Item Drop Emergency Use Case. This feature allows the Citizen to input their contact details and initiate an Alert Staff notification when a valuable item is accidentally dropped, enabling the Cleaning Personnel to assist with retrieval (FR8.0).

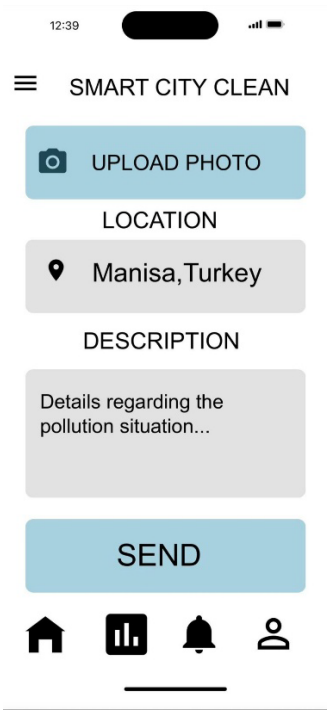
Information/Status: Sections like “City Cleanliness Status” and “Nearby Smart Bins” provide the Citizen with transparent information regarding the general hygiene score of their area and the real-time location of system assets (FR5.0).

https://pr.to/COQ5D9/


20:09
SMART CITY CLEAN
E-MAIL
PASSWORD
ENTER
SIGN UP


20:09
SMART CITY CLEAN
SUBMIT REPORT
NEARBY SMART BINS
CITY CLEANESS STATUS
%82


20:09
SMART CITY CLEAN
BIN FULLNESS
A bin is 80% full
80% Full
A bin is currently 80% full
ISTANBUL 65%
ANKARA 55%
IZMIR 40%
BURSA 35%


12:39
SMART CITY CLEAN
UPLOAD PHOTO
LOCATION
Manisa,Turkey
DESCRIPTION
Details regarding the pollution situation...
SEND


13:50
SMART CITY CLEAN
SEARCH HERE
NEARBY SMART BINS
BİN:Kadıköy-#bin_01
%78 FULL
İstanbul
Ankara
İzmir

4 Glossary

TERM	DEFINITION
Smart Waste Bin	A physical unit with 5 compartments (Metal, Paper, Plastic, Glass, Household Waste), equipped with real-time fullness level and waste classification sensors. It transmits data to the central system.
System	The main software platform that collects and processes data from Smart Waste Bins, performs route optimization, and ensures communication with all actors.
Cleaning Personnel	A team member who receives field tasks via the mobile application, empties the waste bins, and updates task statuses.
Municipal Manager	The high-level actor who monitors the general city cleanliness levels and resource allocation via a data visualization dashboard.
Citizen	The end-user who reports pollution/overflowing bins via the mobile application and initiates an emergency notification when necessary.
Fullness Threshold	A predefined level (e.g., 80%) that, when exceeded by a compartment's fullness level, automatically triggers the System to send an alert to the Cleaning Personnel and Municipal Manager.
Route Optimization	The algorithm that calculates the most efficient collection routes based on collected fullness level data and Citizen reports.
Emergency Emptying Alert	The notification automatically sent by the System to the Cleaning Personnel and Municipal Manager when the fullness threshold is exceeded.
Waste Classification	The process by which the Smart Waste Bin uses sensor data to determine the type of waste deposited (Metal, Paper, Plastic, etc.). It provides an audio/visual warning if incorrect waste is deposited.
Critical Drop	The situation where a Citizen accidentally drops a valuable item (e.g., a wallet) into the Smart Waste Bin. The System alerts the Personnel for item retrieval.
Mobile Application	The interface where the Cleaning Personnel views their optimized routes and confirms task completion, and where the Citizen can submit pollution/overflow reports.

5 Appendix

- Annex – I: Distribution of Work
- Annex – II: Meeting Minutes

Annex – I

Distribution of Work

• Aleyna GÜLTEKİN: Was responsible for preparing the high-level Scenarios (brief flow of events).

• Esra SEPİK: Contributed to writing the detailed Use Case Descriptions / Textual Descriptions (specifically the tables defining the actors, conditions, and flow of events).

• Elif Sude YILMAZ: Handled the creation of the foundational Introduction and Current System sections.

• Elif ÖZKANAT: Focused on the requirements structure, writing the Functional Requirements table and the Nonfunctional Requirements section (Performance, Reliability, Usability, etc.).

• Naciye Nur AKKUŞ: Was tasked with generating and ensuring the accuracy of the Use Case Diagrams (UML diagrams drawn with Draw.io).

• Sinem TANRIVER: Was fully responsible for the Glossary section and preparing the User Interface Mock-ups.

Annex – II

Meeting Minutes

<Copy this sheet as much as you need (for each meeting)>

Date:	16/10/2025
Location:	Engineering Faculty
Duration:	3 Hours
Participants:	Aleyna Gültekin, Sinem Tanrıver, Elif Özkanat, Esra Sepik, Naciye Nur Akkuş, Elif Sude Yılmaz

Content of the meeting (briefly explain the agenda, decisions, work distributions, etc.)

Introduction and Context Formation:

• The Introduction section of the project has been written, defining the purpose of the document and the motivation behind the project.

• The Current System section has been completed, summarizing the issues of traditional waste collection systems and the limited scope of existing technological pilot projects (IoT, basic route planning).

2. Requirements Definition:

• The Proposed System has been described as an integrated and data-driven platform.

• A Functional Requirements table (FR1.0 - FR9.0) outlining the core functionalities of the system has been created.

• A list of Nonfunctional Requirements (Performance, Reliability, Usability, Security, etc.) has been written to define the system's quality and sustainability standards.

Date:	23/10/2025
Location:	Engineering Faculty
Duration:	2 Hours
Participants:	Aleyna Gültekin, Sinem Tanrıver, Elif Özkanat, Esra Sepik, Naciye Nur Akkuş, Elif Sude Yılmaz

Content of the meeting (briefly explain the agenda, decisions, work distributions, etc.)

On the second day, the project's scenarios and Use Case Model were developed. Three main scenarios were defined: a citizen accidentally dropping their wallet into a smart bin and retrieving it with staff assistance, the system giving audio-visual warnings when incorrect waste disposal is attempted, and the system performing route optimization and notifying cleaning staff when a bin exceeds 80% capacity. These scenarios illustrate the interactions among the citizen, cleaning staff, and municipal manager. The developed application was designed according to these models, including modules for citizens to report and send notifications, for cleaning staff to manage routes and tasks, and for managers to monitor bin fullness levels and performance data.

Annex – II

Date:	05/11/2025
Location:	Engineering Faculty
Duration:	2 hours
Participants:	Aleyna Gültekin, Sinem Tanrıver, Elif Özkanat, Esra Sepik, Naciye Nur Akkuş, Elif Sude Yılmaz

Content of the meeting (briefly explain the agenda, decisions, work distributions, etc.)

On the third day, the team focused on completing the Mock-up Design, Glossary, and Work Distribution sections of the "Smart City Clean" project. The mock-ups were designed to visually represent the user interfaces of the system, including the citizen mobile app, cleaning staff application, and municipal manager dashboard, ensuring each interface matched the previously developed use cases. The Glossary section was prepared to define the technical and domain-specific terms (such as IoT sensors, route optimization, and smart bin) to maintain clarity and consistency throughout the document. Finally, the Work Distribution part detailed the balanced contribution of each team member during this project phase, explaining how tasks like documentation, mock-up creation, and scenario validation were shared across the group.